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PhD Chemical Physics
Institute des Sciences Moléculaires
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EXPERIENCE

- Postdoctoral Contract Institute des Sciences Moléculaires, Université de Bordeaux, Jan 2025–Dec 2025, Bordeaux, France
- Doctoral Contract Institute des Sciences Moléculaires, Université de Bordeaux Oct 2021 – Dec 2024 Bordeaux, France
- Teaching Assistant, Atomic and Molecular Physics Department Instituto Superior de Tecnologías y Ciencias Aplicadas, Universidad de La Habana, September 2017 – 2021, La Habana, Cuba

- Development of Computational methods for colliding rovibrational excited water molecules with atoms or rotating linear molecules.
- Scattering calculations for water-atom and water-H₂ systems.
- Construction of Potential Energy Surfaces.
- Calculation of Bound States.

VISIT TO FOREIGN INSTITUTIONS

Universidad Autónoma de Chile

September 2018 – October 2018 Santiago, Chile

Universidad Autónoma de Chile

October 2019 – November 2019 Santiago, Chile

POSTGRAD COURSES

- Ultracold molecules: Quantum physics and applications École de Physique des Houches, 2023, Les Houches, France
- Scattering Theory Summer School, Gdańsk University of Technology, 2022, Gdańsk, Poland
- Methods for Electronic Structure Calculations in Materials, Universidad de La Habana, Instituto de Ciencia y Tecnología de Materiales, 2020, La Habana, Cuba
- Functional Analysis and PDEs, Universidad de La Habana, Facultad de Matemáticas y Computación, 2019, La Habana, Cuba
- Workshop on Parallel and Distributed Computing, Instituto de Cibernéticas, Matemáticas y Física, 2018, La Habana, Cuba
- II International Astrochemistry School, Universidad Autónoma de Chile, October 2018, Santiago, Chile

RESEARCH'S PARTICIPATION

- Participation in the Project: ANR JET3DINBAR, (Jan. 2025-Ongoing), Bordeaux, France
- Participation in the Project: ANR Waterstars, (2021-Ongoing), Bordeaux, France
- Participation in the Project: ECOS-SUD-ANID project number C22E02, (2022-Ongoing), Bordeaux, France
- Participation in the Project: CONICYT/FONDECYT /REGULAR/No.1200732, (2018-2019), Santiago, Chile
- Participation in the Project: No.p223LH001-108 (2017-2020), Havana, Cuba

EDUCATION

- PhD Chemical Physics, Université de Bordeaux, Oct 2021-Dec 2024, Bordeaux, France
- B.E. in Nuclear Technologies and Energetic Installations Instituto Superior de Tecnologías y Ciencias Aplicadas, Universidad de La Habana Sept 2012 - July 2017

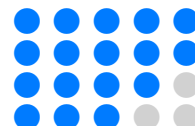
LANGUAGES

Spanish
English
French



PROGRAMMING SKILLS

Fortran
Python
Bash
C++



SOFTWARE & TOOLS

LaTeX

Molpro

Office

Data Visualization: gnuplot, Matplotlib

Data Handling & Analysis: numpy, pandas, scipy

REFEREES

Dr. Thierry Stoecklin

@ Institut des Sciences Moléculaires, Université de Bordeaux

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Prof. Jesús Rubayo Soneira

@ Instituto Superior de Tecnologías y Ciencias Aplicadas, Universidad de La Habana

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Ave. Salvador Allende No. 1110 e/ Infanta y Rancho Boyeros, Plaza de la Revolución, La Habana, Cuba

PUBLICATIONS

Citations: 45

h-index: 4

i10-index: 2

Journal articles

- R. M. García-Vázquez, Z.-F. Sun, C.-H. Yang, *et al.*, "A comprehensive study of the differential cross sections for water-rare gas collisions: Experimental and theoretical perspectives," *Phys. Chem. Chem. Phys.*, pp. -, 2025. DOI: 10.1039/D4CP04825C.
- M. L. Dubernet, C. Boursier, O. Denis-Alpizar, *et al.*, "Basecol2023 scientific content," *Astron. Astrophys.*, vol. 683, A40, Mar. 2024, ISSN: 1432-0746. DOI: 10.1051/0004-6361/202348233.
- R. M. García-Vázquez, A. Bergeat, O. Denis-Alpizar, A. Faure, T. Stoecklin, and S. B. Morales, "Scattering resonances in the rotational excitation of hdo by ne and normal-h2: Theory and experiment," *Faraday Discussions*, vol. 251, pp. 205–224, 2024, ISSN: 1364-5498. DOI: 10.1039/d3fd00168g.
- R. M. García-Vázquez, L. D. Cabrera-González, O. Denis-Alpizar, and T. Stoecklin, "Vibrational relaxation of d2o induced by collision with he: A rigid bender close coupling study," *ChemPhysChem*, vol. 25, no. 15, Jun. 2024, ISSN: 1439-7641. DOI: 10.1002/cphc.202400353.
- R. M. García-Vázquez, L. D. Cabrera-González, O. Denis-Alpizar, and T. Stoecklin, "A rigid bender study of the bending relaxation of h2o and d2o by collisions with ar," *ChemPhysChem*, vol. 25, no. 10, Mar. 2024, ISSN: 1439-7641. DOI: 10.1002/cphc.202300752.
- L. D. Cabrera-González, R. M. García-Vázquez, D. Páez-Hernández, T. Stoecklin, and O. Denis-Alpizar, "Rotational relaxation of sicsi by collision with para-h₂(*j* = 0)," *The European Physical Journal D*, vol. 77, no. 6, Jun. 2023, ISSN: 1434-6079. DOI: 10.1140/epjd/s10053-023-00690-w.
- R. M. García-Vázquez, O. Denis-Alpizar, and T. Stoecklin, "A comparative study of the cold collisions of h2o and d2o with ne," *The Journal of Physical Chemistry A*, vol. 127, no. 22, pp. 4838–4847, 2023, PMID: 37222149. DOI: 10.1021/acs.jpca.3c02086. eprint: <https://doi.org/10.1021/acs.jpca.3c02086>.
- R. M. García-Vázquez, A. Faure, and T. Stoecklin, "Bending relaxation of h2o by collision with para- and ortho-h2," *ChemPhysChem*, vol. 25, no. 2, Dec. 2023, ISSN: 1439-7641. DOI: 10.1002/cphc.202300698.
- R. M. García-Vázquez, M. Márquez-Mijares, J. Rubayo-Soneira, and O. Denis-Alpizar, "Relaxation of ArH⁺ by collision with He: Isotopic effects," *Astron. Astrophys.*, vol. 631, A86, A86, Nov. 2019. DOI: 10.1051/0004-6361/201935712.
- R. M. García-Vázquez, L. Barrios, M. Márquez-Mijares, and T. González-Lezana, "Modelo estadístico cuántico para el estudio de las reacciones átomo-diátomo.," *Revista Cubana de Física*, 2017.

CONGRESS & SYMPOSIUMS

XII International meeting on Photodynamics and Related Aspect. Talk

 [Full Dimensional Quantum Approach to the Vibrational Relaxation of H₂O by Collision with H₂](#)

 R. M. García-Vázquez, Alexandre Faure and T. Stoecklin

 November 2024

 Santiago, Chile

Journée de l'Ecole Doctorale des Sciences Chimiques. Talk

 [Bending Relaxation of H₂O by collision with para- and ortho-H₂](#)

 R. M. García-Vázquez, A. Faure, T. Stoecklin

 March 2024

 Talence, France

Faraday Discussion: New Directions in Molecular Scattering. Poster

 [Inelastic collision of H₂O, HDO and D₂O with rare gases: propensity rules and comparison with experiments.](#)

 R. M. García-Vázquez, A. Bergeat, L.D. Cabrera, O. Denis-Alpizar, T. Stoecklin and S. B. Morales.

 May 2024

 Edinburgh, UK

ISM Day. Talk

 [A Comparative Study of the Inelastic Collisions of H₂O, HDO and D₂O with Ne](#)

 R. M. García-Vázquez, A. Bergeat, O. Denis-Alpizar, T. Stoecklin

 April 2023

 Talence, France

XI International meeting on Photodynamics and Related Aspect. Poster

 [Study of the bending relaxation of water by collision with Ar.](#)

 R. M. García-Vázquez, L.D. Cabrera-González, O. Denis-Alpizar and T. Stoecklin

 November 2022

 La Habana, Cuba

JTMS 2022: Journées "Théorie, Modélisation et Simulation". Poster

 [Study of the bending relaxation of water by collision with Ar using the Rigid Bender Close Coupling treatment](#)

 R. M. García-Vázquez, L.D. Cabrera-González, O. Denis-Alpizar and T. Stoecklin

 October 2022

 Rennes, France

Rencontres des Chimistes Théoriciens Francophones. Poster

 [Études préliminaires des collisions inélastiques de l'eau avec l'argon](#)

 R. M. García-Vázquez, L.D. Cabrera-González, O. Denis-Alpizar and T. Stoecklin

 July 2022

 Bordeaux, France

LatinXChem. Poster.

 [Potential Energy Surfaces for the triplet ground states of ArHC⁺.](#)

 R. M. García-Vázquez, M. Márquez-Mijares, R. Urzua Leiva, J. Rubayo-Soneira and O. Denis-Alpizar

 2020

 Online Congress, via Twitter.

 [certificate](#)

XV Cuban Physics Society Symposium. Poster

 [Isotopic Effects Study of the Inelastic Collision and Vibrational Predissociation on the ArH⁺ – He system.](#)

 R. M. García-Vázquez, M. Márquez-Mijares, J. Rubayo-Soneira, O. Denis-Alpizar

 2020

 Havana, Cuba.

 [certificate](#)

II Physic Meeting for postgrad student. Poster

 [Relaxation of ArH⁺ by collision with He: Isotopic effects](#)

 R. M. García-Vázquez, M. Márquez-Mijares, J. Rubayo-Soneira, O. Denis-Alpizar

 2020

 Havana, Cuba.

 [certificate](#)

WorkShop "Understanding Molecular Dynamics: Theory and Applications"

 R. M. García-Vázquez, M. Márquez-Mijares, J. Rubayo-Soneira, O. Denis-Alpizar

 2018

 Havana, Cuba

 [certificate](#)

XIV Cuban Physics Society Symposium. Poster

 [Modelo Estadístico Cuántico para el estudio de las reacciones Si+OH y O+H₂](#)

 R. García-Vázquez, L. Barrios, M. Márquez-Mijares y T. González-Lezana

 2017

 Havana, Cuba.

 [certificate](#)

Eighth International Symposium on Atomic Cluster Collisions (ISACC 2017), Poster

 [Approximations to the Statistical Quantum Model for atom-diatom reaction studies](#)

 R. García-Vázquez, L. Barrios, M. Márquez-Mijares y T. González-Lezana

 2017

 Varadero, Matanzas, Cuba.

 [certificate](#)